

RAPID (1/3)

Research Objectives

- Few-Step generating Diffusion
- Data and step adaptive prior in Diffusion

Motivation

- Problem Definition
 - Too slow sampling in Diffusion.
 - Diffusion's capacity is wasted in early time steps.
- Core Causes
 - Gaussian prior carries zero information
 - → model must learn structure from scratch.
 - → curved, entangled velocity trajectory.
 - → required many sampling steps.
- Related Works
 - Existing work optimizes the trajectory, not the inefficient prior that causes it.

Topic: Few-Step Diffusion via Adaptive Prior

Title: RAPID: Robust Adaptive Prior Integration Diffusion

What is **RAPID**?

- Diagnoses sampling inefficiency at its source
 - The prior, not the trajectory
- Structured GMM prior, principled integration into diffusion
- Coarse-to-fine design → high-quality few-step generation
- **RAPID** = Robust Adaptive Prior Integration for Diffusion

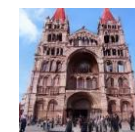
Naïve Diffusion

250 steps, 0.63 img/s



RAPID

40 steps, 3.99 img/s



t = 0.0

t = 0.5

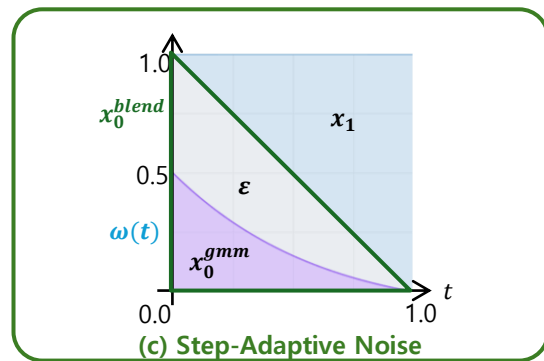
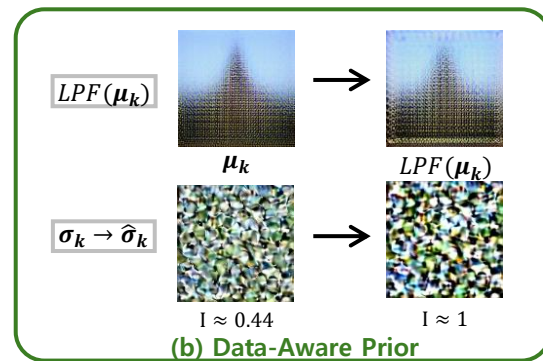
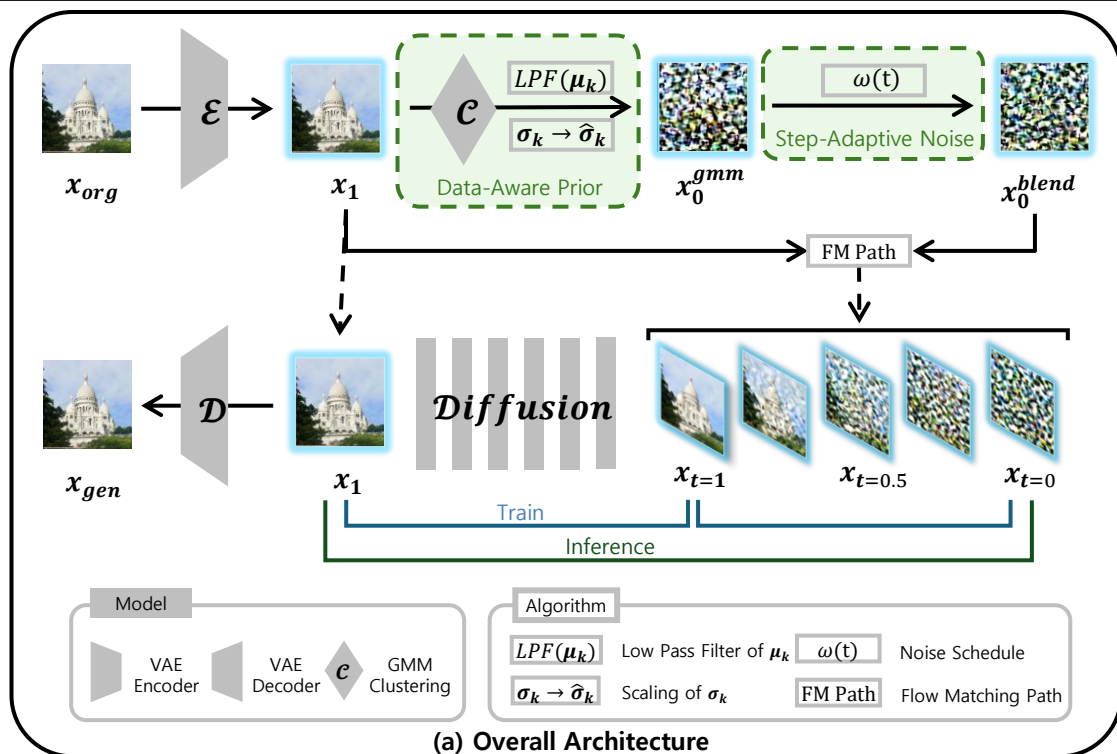
t = 0.75

t = 1.0

RAPID is 6.3x faster! **RAPID** is 84% fewer steps!

RAPID (2/3)

Model Architecture



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Method

➤ Data-Aware Prior

- GMM clustering of latent features
→ reveals structural modes already in target dataset.
- Cluster mean low-pass filter (LPF)
→ keeps coarse structure only.
- Cluster covariance normalization
→ restores diversity for diffusion's stochastic.

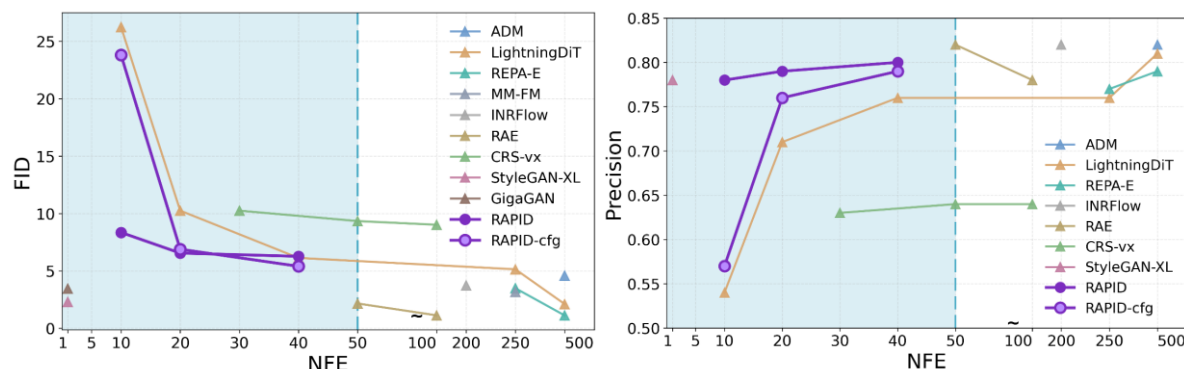
➤ Step-Adaptive Noise

- Coarse-to-fine principle
→ GMM prior's role should fade over time.
- SNR-aware blending $w(t)$
→ controls this decay.
- High $w(t)$ early (structure) → low $w(t)$ late (fine detail)

Experimental Results

Quantitative Results

- Strong performance preserved at 20 to 40 NFE(= few sampling step).
- 40 NFE: FID 5.40 on ImageNet-1K, trained for 50 epochs



Qualitative



Achievement

- The adaptive prior approach underlying this thesis was developed through two preceding studies (First author).

- "An Optimized Diffusion Model Based on Adaptive Prior Distribution," KMMS Autumn Conference, Nov. 2025. (Best Paper Award)
- "Adaptive Prior Diffusion: Dataset-Aware GMM Prior for High-Fidelity Image Generation," KMMS Spring Conference, May 2026. (Best Paper Award)

- The current thesis work is being prepared for submission to *Pattern Recognition*.